

Table 7.2-1 (Sheet 1 of 2)
Reactor Trip Variables, Limits, Ranges, and Accuracies
(Design Basis for Reactor Trip)
(Nominal)

Protective Functions	Variable	Range of Variables	Typical Accuracy⁽¹⁾	Typical Response Time (Sec)⁽²⁾
Source Range High Neutron Flux	Neutron flux	6 decades of neutron flux: 1 to 10 ⁶ counts per second	±10% of span	0.6
Intermediate Range High Neutron Flux	Neutron flux	8 decades of neutron flux overlapping source range by 2 decades and including 100% power	±10% of span	0.6
Power Range Neutron Flux (Low Setpoint)	Neutron flux	1 to 120% of full power	±5% of span	0.6
Power Range Neutron Flux (High-Setpoint)	Neutron flux	1 to 120% of full power	±5% of span	0.6
Power Range High Positive Flux Rate	Neutron flux	1 to 120% of full power	±1% of span	0.6 (step input of 20% full power)
Overtemperature ΔT			±5% of ΔT span	
	Cold leg temp. (T _{cold})	490° to 610°F		5.5
	Hot leg temp. (T _{hot})	530° to 650°F		5.5
	Pressurizer pressure	1700 to 2500 psig	±3% of span	0.9
	Neutron flux (difference between top and bottom power range detectors)	-60 to +60% (Δφ)		0.6
Overpower ΔT			±4% of ΔT span	
	Cold leg temp. (T _{cold})	490° to 610°F		5.5
	Hot leg temp. (T _{hot})	530° to 650°F		5.5
	Pressurizer pressure	1700 to 2500 psig	±3% of span	0.9
	Neutron flux (difference between top and bottom power range detectors)	-60 to +60% (Δφ)		0.6

Table 7.2-1 (Sheet 2 of 2)
Reactor Trip Variables, Limits, Ranges, and Accuracies
(Design Basis for Reactor Trip)
(Nominal)

Protective Functions	Variable	Range of Variables	Typical Accuracy⁽¹⁾	Typical Response Time (Sec)⁽²⁾
Pressurizer Low Pressure	Pressurizer pressure	1700 to 2500 psig	±3% of span	0.9
Pressurizer High Pressure	Pressurizer pressure	1700 to 2500 psig	±3% of span	0.9
Pressurizer High Water Level	Pressurizer water level	0-100% of entire cylindrical portion of pressurizer	±5% of span	0.9
Low Reactor Coolant Flow	Coolant flow	0 to 120% of rated flow	±3% of span	0.9
Low Reactor Coolant Pump Speed	Pump speed	0 to 120% of rated speed	±1% of span	0.7
Low Steam Generator Water Level	Steam generator water level	0-100% of span (narrow range taps)	±22% of span	0.9
High Steam Generator Water Level	Steam generator water level	0-100% of span (narrow range taps)	±13% of span	0.9
Reactor Coolant Pump High Bearing Water Temperature	Reactor coolant pump bearing water temperature	70°-450°F	±2% of span	5.5
Automatic or Manual Safeguards Actuation	See Table 7.3-4	See Table 7.3-4	See Table 7.3-4	See Table 7.3-4
Manual Reactor Trip	Control status	N/A	N/A	N/A
Automatic or Manual Depressurization System Actuation	See Table 7.3-4	See Table 7.3-4	See Table 7.3-4	See Table 7.3-4
Automatic or Manual Core Makeup Tank Injection	See Table 7.3-4	See Table 7.3-4	See Table 7.3-4	See Table 7.3-4
Automatic or Manual PRHR Actuation	PRHR discharge valve position	Valve closed/valve not-closed	N/A	1.25

Notes:

1. Measurement uncertainty typical of actual applications. Harsh environment allowances have been included where applicable.
2. Delay from the time that the process variable exceeds the setpoint until the time that the control rods are free to fall into the core (includes reactor trip breaker opening delay and control rod drive mechanism gripper release delay).

**Table 7.2-2
Reactor Trips**

Reactor Trip⁽¹⁾	No. of Channels	Division Trip Logic	Bypass Logic	Permissives and Interlocks (See Table 7.2-3)
Source Range High Neutron Flux Reactor Trip	4	2/4	Yes ⁽²⁾	P-6, P-10
Intermediate Range High Neutron Flux Reactor Trip	4	2/4	Yes ⁽²⁾	P-10
Power Range High Neutron Flux (Low Setpoint) Trip	4	2/4	Yes ⁽²⁾	P-10
Power Range High Neutron Flux (High Setpoint) Trip	4	2/4	Yes ⁽²⁾	----
High Positive Flux Rate Trip	4	2/4	Yes ⁽²⁾	----
Reactor Coolant Pump Bearing Water Temperature	16 (4/pump)	2/4 in any single pump	Yes ⁽²⁾	----
Overtemperature ΔT	4 (2/loop)	2/4	Yes ⁽²⁾	----
Overpower ΔT	4 (2/loop)	2/4	Yes ⁽²⁾	----
Pressurizer Low Pressure Trip	4	2/4	Yes ⁽²⁾	P-10
Pressurizer High Pressure Trip	4	2/4	Yes ⁽²⁾	----
High-3 Pressurizer Water Level Trip	4	2/4	Yes ⁽²⁾	P-10
Low Reactor Coolant Flow	8 (4/hot leg)	2/4 in either hot leg	Yes ⁽²⁾	P-10
Reactor Coolant Pump Underspeed	4 (1/pump)	2/4	Yes ⁽²⁾	P-10
Low Steam Generator Water Level	4/steam generator	2/4 in any steam generator	Yes ⁽²⁾	----
High-2 Steam Generator Water Level	4/steam generator	2/4 in any steam generator	Yes ⁽²⁾	P-11
Automatic Safeguards Actuation	4	2/4	Yes ⁽²⁾	—
Automatic Depressurization System Actuation	4	2/4	Yes ⁽²⁾	—
Automatic Core Makeup Tank Injection	4	2/4	Yes ⁽²⁾	—
PRHR Actuation	4	2/4	Yes ⁽²⁾	—
Manual Safeguards Actuation	2 controls	1/2 controls	No	—
Manual Depressurization System Actuation	4 controls	2/4 controls	No	—
Manual Core Makeup Tank Injection	2 controls	1/2 controls	No	—
Manual Reactor Trip	2 controls	1/2 controls	No	—

Notes:

- Reactor Trip divisions are also bypassed with the logic as defined in 2. below.
 - Bypass Logic = 2/4 with no bypasses; 2/3 with 1 bypass; more than one bypass is not allowed.
- No permissive or interlock.

Table 7.3-1 (Sheet 1 of 7)
Engineered Safety Features Actuation Signals

Actuation Signal	No. of Divisions/ Controls	Actuation Logic	Permissives and Interlocks
1. Safeguards Actuation Signal (Figure 7.2-1, Sheets 9 and 11)			
a. Low pressurizer pressure	4	2/4-BYP ¹	Can be manually blocked on presence of P-3 Block automatically removed on absence of P-3 Manual block permitted below P-11 Automatically unblocked above P-11
b. Low lead-lag compensated steam line pressure	4/steam line	2/4-BYP ¹ in either steam line	Can be manually blocked on presence of P-3 Block automatically removed on absence of P-3 Manual block permitted below P-11 Automatically unblocked above P-11
c. Low cold leg temperature (Low T _{cold})	4/loop	2/4-BYP ¹ either loop ⁶	Can be manually blocked on presence of P-3 Block automatically removed on absence of P-3 Manual block permitted below P-11 Automatically unblocked above P-11
d. High-2 containment pressure	4	2/4-BYP ¹	Can be manually blocked on presence of P-3 Block automatically removed on absence of P-3
e. Manual safeguards initiation	2 controls	1/2 controls	None
2. Containment Isolation (Figure 7.2-1 Sheets 11 and 13)			
a. Automatic or manual safeguards actuation signal	(See items 1a through 1e)		
b. Manual initiation	2 controls	1/2 controls	None
c. Manual initiation of passive containment cooling	(See item 10a)		
3. Automatic Depressurization System (Figure 7.2-1, Sheet 15) (Initiate Stages 1, 2, and 3)			
a. Core makeup tank injection coincident with	(See items 6a through 6e)		
Core makeup tank level less than Low-1 setpoint	4/tank	2/4-BYP ¹ either tank ²	None
b. Extended undervoltage to Class 1E battery chargers ⁽⁸⁾	2/charger	1/2 per charger and 2/4 chargers	None
c. Stages 1, 2, and 3 manual initiation	4 controls	2/4 controls ³	None

Table 7.3-1 (Sheet 2 of 7)
Engineered Safety Features Actuation Signals

Actuation Signal	No. of Divisions/ Controls	Actuation Logic	Permissives and Interlocks
(Initiate Stage 4)			
d. Stage 4 manual initiation coincident with one of the following two conditions:	4 controls	2/4 controls ³	None
Low reactor coolant system pressure or	4	2/4 BYP ¹	None
3rd stage depressurization			
e. Core makeup tank level less than Low-2 setpoint coincident with	4/tank	2/4 BYP ¹ either tank ²	None
Low reactor coolant system pressure and coincident with	4	2/4 BYP ¹	None
3rd stage depressurization			
f. Coincident loop 1 and loop 2 Low-2 hot leg level (after delay)	1 per loop	2/2	Manual unblock permitted below P-12 Automatically blocked above P-12
4. Main Feedwater Isolation (Figure 7.2-1, Sheet 10) (Closure of Control Valves)			
a. Safeguards actuation signal (automatic or manual)	(See items 1a through 1e)		
b. Manual initiation	2 controls	1/2 controls	None
c. High-2 steam generator narrow range level	4/steam generator	2/4-BYP ¹ in either steam generator	None
d. Low reactor coolant temperature (Low-1 T _{avg}) coincident with	2/loop	2/4 -BYP ¹	Manual block permitted below P-11 Automatically unblocked above P-11
Reactor trip (P-4)	1/division	2/4	None
(Trip of Main Feedwater Pumps and Closure of Isolation and Crossover Valves)			
a. Safeguards actuation signal (automatic or manual)	(See items 1a through 1e)		
b. Manual initiation	2 controls	1/2 controls	None
c. High-2 steam generator narrow range level	4/steam generator	2/4-BYP ¹ in either steam generator	None
d. Low reactor coolant temperature (Low-2 T _{avg}) coincident with	2/loop	2/4-BYP ¹	Manual block permitted below P-11 Automatically unblocked above P-11
Reactor trip (P-4)	1/division	2/4	None

Table 7.3-1 (Sheet 3 of 7)
Engineered Safety Features Actuation Signals

Actuation Signal	No. of Divisions/ Controls	Actuation Logic	Permissives and Interlocks
5. Reactor Coolant Pump Trip (Figure 7.2-1, Sheets 5, 7, 12, and 15) (Trips All Reactor Coolant Pumps)			
a. Safeguards actuation signal (automatic or manual)	(See items 1a through 1e)		
b. Automatic reactor coolant system depressurization (first stage)	(See items 3a through 3c)		
c. Low-2 pressurizer level	4	2/4-BYP ¹	Manual block permitted below P-12 Automatically unblocked above P-12
d. Low wide range steam generator water level coincident with	4/steam generator	2/4-BYP ¹ in both steam generators	None
High hot leg temperature (High T _{hot}) ⁽⁸⁾	2/loop	2/4-BYP ¹	None
e. Manual core makeup tank initiation	(See item 6e)		
f. High reactor coolant pump bearing water temperature	4/pump	2/4-BYP ¹ in affected pump	None
6. Core Makeup Tank Injection (Figure 7.2-1, Sheets 7, 12 and 15)			
a. Safeguards actuation signal (automatic or manual)	(See items 1a through 1e)		
b. Automatic reactor coolant system depressurization (first stage)	(See items 3a through 3c)		
c. Low-2 pressurizer level	4	2/4-BYP ¹	Manual block permitted below P-12 Automatically unblocked above P-12
d. Low wide range steam generator water level coincident with	4/steam generator	2/4-BYP ¹ in both steam generators	None
High hot leg temperature (High T _{hot}) ⁽⁸⁾	2/loop	2/4-BYP ¹	None
e. Manual initiation	2 controls	1/2 controls	None
7. Turbine Trip (Figure 7.2-1, Sheet 14)			
a. Manual feedwater isolation	(See item 4b)		
b. Reactor trip (P-4)	1/division	2/4	None
c. High-2 steam generator narrow range level	4/steam generator	2/4-BYP ¹ in either steam generator	None
8. Steam Line Isolation (Figure 7.2-1, Sheet 9)			
a. Manual initiation	2 controls	1/2 controls	None
b. High-2 containment pressure	4	2/4-BYP ¹	None
c. Low lead-lag compensated steam line pressure ⁴	4/steam line	2/4-BYP ¹ in either steam line	Manual block permitted below P-11 Automatically unblocked above P-11

Table 7.3-1 (Sheet 4 of 7)
Engineered Safety Features Actuation Signals

Actuation Signal	No. of Divisions/ Controls	Actuation Logic	Permissives and Interlocks
d. High steam line negative pressure rate	4/steam line	2/4-BYP ¹ in either steam line ⁷	Manual unblock permitted below P-11 Automatically blocked above P-11
e. Low cold leg temperature (Low T _{cold})	4/loop	2/4-BYP ¹ either loop ⁶	Manual block permitted below P-11 Automatically unblocked above P-11
9. Steam Generator Blowdown System Isolation (Figure 7.2-1 Sheets 7 and 8)			
a. Passive residual heat removal heat exchanger actuation	(See items 12a through 12f)		
b. Low narrow range steam generator water level	4/steam generator	2/4 BYP ¹ in either steam generator	None
10. Passive Containment Cooling Actuation (Figure 7.2-1, Sheet 13)			
a. Manual initiation	2 controls	1/2 controls	None
b. High-2 containment pressure	4	2/4-BYP ¹	None
11. Startup Feedwater Isolation (Figure 7.2-1, Sheets 9 and 10)			
a. Low cold leg temperature (Low T _{cold})	4/loop	2/4-BYP ¹ either loop ⁶	Manual block permitted below P-11 Automatically unblocked above P-11
b. High-2 steam generator narrow range water level	4/steam generator	2/4-BYP ¹ in either steam generator	None
c. Manual initiation of main feedwater isolation		(See item 4b)	
d. High steam generator narrow range level coincident with	4/steam generator	2/4-BYP ¹ in either steam generator	None
Reactor trip (P-4)	1/division	2/4	None
12. Passive Residual Heat Removal (Figure 7.2-1, Sheet 8)			
a. Manual initiation	2 controls	1/2 controls	None
b. Low narrow range steam generator water level coincident with	4/steam generator	2/4-BYP ¹ in either steam generator	None
Low startup feedwater flow	2/feedwater line	1/2 in either feedwater line	None
c. Low steam generator wide range water level	4/steam generator	2/4-BYP ¹ in either steam generator	None
d. Core makeup tank injection	(See Items 6a through 6e)		
e. Automatic reactor coolant system depressurization (first stage)	(See items 3a through 3c)		
f. High-3 pressurizer level	4	2/4-BYP ¹	Manual block permitted below P-19 Automatically unblocked above P-19

Table 7.3-1 (Sheet 5 of 7)
Engineered Safety Features Actuation Signals

Actuation Signal	No. of Divisions/ Controls	Actuation Logic	Permissives and Interlocks
13. Block of Boron Dilution (Figure 7.2-1, Sheets 3 and 15)			
a. Flux doubling calculation	4	2/4-BYP ¹	Manual block permitted above P-8 Automatically unblocked (momentary) below P-6 or below P-8 Demineralized water system isolation valves closed if blocked below P-8
b. Undervoltage to Class 1E battery chargers ⁽⁸⁾	2/charger	2/2 per charger and 2/4 chargers ⁵	None
c. Reactor trip (P-4)	1/division	2/4	None
14. Chemical Volume Control System Isolation (See Figure 7.2-1, Sheets 6 and 11)			
a. High-2 pressurizer water level	4	2/4-BYP ¹	Automatically unblocked above P-19 Manual block permitted below P-19
b. High-2 steam generator narrow range level	4/steam generator	2/4-BYP ¹ in either steam generator	None
c. Automatic or manual safeguards actuation signal coincident with	(See items 1a through 1e)		
High-1 pressurizer water level	4	2/4-BYP ¹	None
d. High-2 containment radioactivity	4	2/4-BYP ¹	None
e. Manual initiation	2 controls	1/2 controls	None
f. Flux doubling calculation	4	2/4-BYP ¹	Manual block permitted above P-8. Automatically unblocked (momentary) below P-6 or below P-8. Demineralized water system isolation valves closed if blocked below P-8
g. High steam generator narrow range level coincident with	4/steam generator	2/4-BYP ¹ in either steam generator	None
Reactor trip (P-4)	1/division	2/4	None
15. Steam Dump Block (Figure 7.2-1, Sheet 10) ⁽⁸⁾			
a. Low reactor coolant temperature (Low-2 T _{avg})	2/loop	2/4-BYP ¹	None
b. Mode control	2 controls	1/division	None
c. Manual stage 1 cooldown control	2 controls	1/division	None
d. Manual stage 2 cooldown control	2 controls	1/division	None

Table 7.3-1 (Sheet 6 of 7)
Engineered Safety Features Actuation Signals

Actuation Signal	No. of Divisions/ Controls	Actuation Logic	Permissives and Interlocks
16. Main Control Room Isolation, Air Supply Initiation, and Electrical Load De-energization (Figure 7.2-1, Sheet 13)			
a. High-2 main control room supply air radiation	2	1/2	None
b. Undervoltage to Class 1E battery chargers ⁽⁸⁾	2/charger	2/2 per charger and 2/4 chargers ⁵	None
c. Manual initiation ⁽⁸⁾	2 controls	1/2 controls	None
17. Auxiliary Spray and Purification Line Isolation (Figure 7.2-1, Sheet 12)			
a. Low-1 pressurizer level	4	2/4-BYP ¹	Manual block permitted below P-12 Automatically unblocked above P-12
b. Manual initiation of chemical and volume control system isolation	(See item 14e)		
c. Manual initiation of auxiliary spray isolation	1	1/1	None
18. Containment Air Filtration System Isolation (Figure 7.2-1, Sheets 11 and 13)			
a. Containment isolation	(See items 2a through 2c)		
b. High-1 containment radioactivity	4	2/4-BYP ¹	None
c. N/A	2	N/A	For containment vacuum relief valves only – close on inside containment purge isolation valve not closed
19. Normal Residual Heat Removal System Isolation (Figure 7.2-1, Sheets 13 and 18)			
a. Automatic or manual safeguards actuation signal	(See items 1a through 1e)		
b. High-2 containment radioactivity	4	2/4-BYP ¹	Manual block permitted below P-11 Automatically unblocked above P-11
c. Manual initiation	4 controls	2/4 controls ³	None
20. Refueling Cavity Isolation (Figure 7.2-1, Sheet 13)			
a. Low spent fuel pool level	3	2/3	None
21. Open In-Containment Refueling Water Storage Tank (IRWST) Injection Line Valves (Figure 7.2-1, Sheets 12 and 16)			
a. Automatic reactor coolant system depressurization (fourth stage)	(See items 3d and 3e)		
b. Manual initiation	4 controls	2/4 controls ³	None
22. Open Containment Recirculation Valves In Series with Check Valves (Figure 7.2-1, Sheet 15 and 16)			
a. Extended undervoltage to Class 1E battery chargers ⁽⁸⁾	2/charger	1/2 per charger and 2/4 chargers	None

Table 7.3-1 (Sheet 7 of 7)
Engineered Safety Features Actuation Signals

Actuation Signal	No. of Divisions/ Controls	Actuation Logic	Permissives and Interlocks
23. Open All Containment Recirculation Valves (Figure 7.2-1, Sheet 16)			
a. Automatic reactor coolant system depressurization (fourth stage) coincident with		(See items 3d through 3f)	
Low IRWST level (Low-3 setpoint)	4	2/4-BYP ¹	None
b. Manual initiation	4 controls	2/4 controls ³	None
24. Chemical and Volume Control System Letdown Isolation (Figure 7.2-1, Sheet 16)			
a. Low-1 hot leg level	1 per loop	1/2	Manual block permitted above P-12 Automatically unblocked below P-12
25. Pressurizer Heater Trip (Figure 7.2-1, Sheets 6 and 12)			
a. Core makeup tank injection		(See items 6a through 6e)	
b. High-3 pressurizer level	4	2/4-BYP ¹	Manual block permitted below P-19 Automatically unblocked above P-19
26. Steam Generator Relief Isolation (Figure 7.2-1, Sheet 9)			
a. Manual initiation	2 controls	1/2 controls	None
b. Low lead-lag compensated steam line pressure ⁴	4/steam line	2/4-BYP ¹ in either steam line	Manual block permitted below P-11 Automatically unblocked above P-11
27. Close Component Cooling System Containment Isolation Valves (Figure 7.2-1, Sheet 5)			
a. High reactor coolant pump bearing water temperature	4/pump	2/4-BYP ¹ in affected pump	None
28. Containment Vacuum Relief (Figure 7.2-1, Sheet 19)			
a. Low-2 containment pressure	4	2/4-BYP ¹	None
b. Manual initiation	2 controls	1/2 controls	None

Notes:

- 2/4-BYP indicates bypass logic. The logic is 2 out of 4 with no bypasses and 2 out of 3 with one bypass.
- Any two channels from either tank not in same division.
- Two associated controls must be actuated simultaneously.
- Also, closes power-operated relief block valve of respective steam generator.
- The two-out-of-four logic is based on undervoltage to the battery chargers for divisions A or C coincident with an undervoltage to the battery chargers for divisions B or D.
- Any two channels from either loop not in same division.
- Any two channels from either line not in same division.
- This function does not meet the 10 CFR 50.36(c)(2)(ii) criteria and is not included in the Technical Specifications.